# Amazon – Global E-Commerce Intelligence Suite – Product Performance Analytics

**Report prepared and analytics conducted by Soham S. Amburle | 23 – 25 February 2026**

The Product Performance Analytics Dashboard provides a strategic and operational analytical view of product-level performance within a global e-commerce business model inspired by Amazon. Built using a synthetic dataset of 100,000 transaction-level records, this dashboard focuses on revenue generation, product profitability, SKU concentration, geographic performance, pricing efficiency, and demand trends.

Designed in Power BI using a structured star schema data model, the dashboard enables executive leadership, category managers, merchandising teams, and commercial stakeholders to evaluate product performance through dynamic filtering and structured visual storytelling.

Through KPI cards, slicers, profitability benchmarking, revenue distribution visuals, trend analysis, geographic mapping, and SKU-level ranking, the dashboard delivers a consolidated and executive-ready perspective of product intelligence and performance.

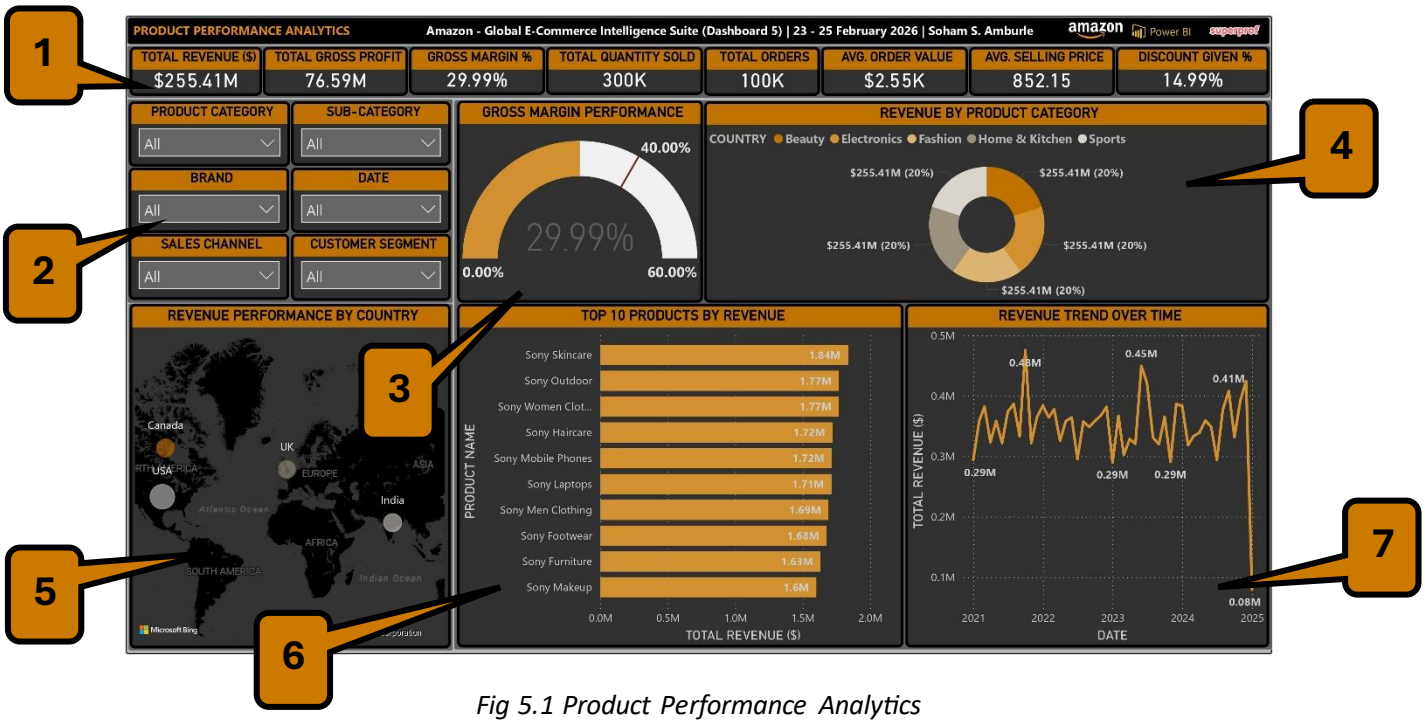


*Fig 5.1 Product Performance Analytics*

## Executive Storyline

The dashboard follows a structured product analytics narrative:

- Establish overall product revenue and profitability scale
- Measure product-level margin efficiency against target benchmarks
- Evaluate revenue distribution across categories
- Identify top-performing SKUs driving revenue concentration
- Analyze geographic revenue distribution
- Assess product demand momentum over time


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This layered structure ensures stakeholders can move from high-level financial metrics to SKU-level and geographic performance insights.

# Visuals and Description

## 1. KPI Cards (Executive Marketing Snapshot)

Eight KPI cards provide an immediate summary of product performance:

1. Total Revenue: Overall revenue generated from products within the selected filter context.
2. Total Gross Profit: Total profit after cost of goods sold.
3. Gross Margin (%): Profitability ratio relative to revenue.
4. Total Quantity Sold: Total units sold across products.
5. Total Orders: Total transactions recorded.
6. Average Order Value (AOV): Revenue generated per order.
7. Average Selling Price (ASP): Revenue generated per unit sold.
8. Average Discount (%): Average discount dependency across product sales.

These KPIs establish product scale, margin health, pricing efficiency, and promotional intensity at an executive level.

## 2. Slicers (Dynamic Filtering Controls)

The dashboard includes six interactive slicers:

1. Product Category
2. Sub Category
3. Brand
4. Date (Between Range Filter)
5. Sales Channel (Web / Mobile App)
6. Customer Segment (Prime / Non-Prime)

All visuals dynamically respond to these filters, enabling multi-dimensional product performance analysis across time, brand, channel, and customer segment.

## 3. Gross Margin vs Target (Gauge)

Value: Gross Margin (%)

Target: 40% Benchmark

This visual highlights:

- Overall product margin efficiency
- Performance against profitability target
- Margin pressure or strength
- Executive-level profitability benchmarking

It provides an immediate indication of product profitability health.

## 4. Revenue Contribution by Category (Donut Chart)

Legend: Product Category

Values: Total Revenue

This visual highlights:


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- Revenue concentration across categories
- Category-level contribution share
- Over-dependence risk on a single category
- Strategic category diversification insights

It complements SKU-level analysis by showing high-level revenue distribution.

## 5. Revenue by Country (Filled Map Visualization)

Location: Customer Country

Color Intensity: Total Revenue

Tooltips include:

- Total Gross Profit
- Gross Margin (%)
- Total Orders

This geographic distribution visual enables:

- Identification of high-performing markets
- Regional product demand comparison
- Geographic revenue concentration analysis
- International performance benchmarking.

## 6. Top 10 Products by Revenue (Clustered Bar Chart)

Axis: Product Name

Values: Total Revenue

Tooltips: Gross Profit, Gross Margin (%), Total Quantity

This visual highlights:

- Revenue-driving SKUs
- Product concentration risk
- High-margin vs high-volume comparison
- Category-level SKU dominance

It enables deep SKU-level performance evaluation.

## 7. Revenue Trend Over Time (Line Chart)

X-Axis: Date (Year-Month)

Y-Axis: Total Revenue

Optional Comparison: Gross Profit

This visual illustrates:

- Product demand momentum
- Seasonal sales fluctuations
- Revenue growth trajectory
- Profitability trend comparison


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It supports strategic forecasting and product lifecycle assessment.

## Business Value

This dashboard enables stakeholders to:

- Monitor product revenue and profitability in a consolidated executive view
- Benchmark margin performance against strategic targets
- Identify top-performing SKUs and category concentration risks
- Evaluate pricing and discount efficiency
- Track geographic revenue distribution
- Assess product demand trends over time

By integrating executive KPIs, profitability benchmarking, SKU ranking, geographic mapping, and trend analytics, the dashboard delivers a structured and enterprise-ready product intelligence framework.

## Tools & Data

- Visualization Tool: Power BI
- Dataset: Fully synthetic global e-commerce dataset
- Record Count: 100,000 transaction-level records
- Time Period: 2021–2024
- Model Design: Star Schema (Fact_Orders connected to Date, Customer, Product, and Channel dimensions)
- Purpose: Product performance analytics and enterprise BI portfolio demonstration


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